

Homework 7

Q. (1) Bigram Model

<S> I study Computer Science </S>

<S> I learn Computer Engineering </S>

<S> We learn Computer Science </S>

<S> We study Computer Science and Computer Engineering </S>

Compute likelihood (probability) of sentence

"I study Computer Science" with the Bigram model

Write down clearly all individual (conditional) probabilities

→ In Bigram probability, we ~~divide~~ calculate the no. of times $w_{i-1} | w_i$ appears in the ~~context~~ given corpus by the total no. of times w_{i-1} appears in the corpus.

Formula

$$P(w_i | w_{i-1}) = \frac{\text{count}(w_{i-1}, w_i)}{\text{count}(w_{i-1})}$$

$$P(I | <S>) = \frac{\text{count}(<S>, I)}{\text{count}(<S>)} = \frac{2}{4} = \frac{1}{2}$$

$$P(\text{study} | I) = \frac{\text{count}(I, \text{study})}{\text{count}(I)} = \frac{1}{2}$$

$$P(\text{Computer} | \text{study}) = \frac{\text{count}(\text{study}, \text{Computer})}{\text{count}(\text{study})} = \frac{2}{2} = 1$$

$$P(\text{Science} | \text{Computer}) = \frac{\text{count}(\text{Computer}, \text{Science})}{\text{count}(\text{Computer})} = \frac{3}{5}$$

$$P(</s> | \text{Science}) = \frac{\text{count}(\text{Science}, </s>)}{\text{count}(\text{Science})} = \frac{2}{3}$$

$$P(\text{I study Computer Science}) = P(\text{I} | <s>) \cdot P(\text{study} | \text{I}) \cdot P(\text{Computer} | \text{study}) \cdot P(\text{Science} | \text{Computer}) \cdot P(</s> | \text{Science})$$

$$= \frac{1}{2} * \frac{1}{2} * 1 * \frac{3}{5} * \frac{2}{3}$$

$$= \frac{1}{10} //$$